

Catchment Water Balance Analysis and Simulation

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December 12, 2024



Notes in regards to the report:

All figures are vectorised, so feel free to zoom in as you please. The code which accompanies this report is well documented, with explanations to what each section does, this is provided in a jupyter notebook named *MajdYousoufCWSsubmission.ipynb*. Please refer to it for more details regarding calculations as they are kept brief due to the page limit.

1 Question 1: Average Water Balance

To start, it is necessary to determine the land cover for the catchment to able to calculate the actual evapotranspiration throughout the time frame. For this, it is assumed that the land cover remains constant along the years, which may present inaccuracies in the analysis. From figure 1, it is clear that the specific catchment is comprised of majority crops/farmland, with grass and pasture as dominant land covers. Exact percentages can be obtained from the [National River Flow Archive \(2024\)](#), and they verify figure 1 which was produced using CAMELS Catchment data ([Coxon et al. 2020](#)).

The next step is to verify the types of crops present in the arable percentage of the catchment, and to do this, the top 3 crops in the southeast of England are identified as wheat, barley and rapeseed; it was assumed that the proportion of the other crops grown are negligible ([Defra 2024](#)). Once this is done, the monthly values of $k_c^{(arable)}$ are calculated and are available in figures 9 and 10 in [Appendix 1: Catchment Water Balance](#) as it changes over time. Now let m represent the different types of land cover, and from this, the $PET_{catchment}$ can be calculated as:

$$PET_{catchment} = PET_i \times \sum_{m=1}^M k_c^{(m)} \times \text{percentage land cover}^{(m)} \quad (1)$$

Where PET_i is the corrected PET to account for interception. The final values for k_c of the catchment can be found in figure 11 in [Appendix 1: Catchment Water Balance](#). Finally, the AET for the catchment can be calculated by multiplying $PET_{catchment}$ by k_{stress} , which is a value calculated using $k_{stress} = \frac{\text{Soil moisture} - \text{Wilting point}}{\text{Field capacity} - \text{Wilting point}}$ and is a value between 0 and 1. Figure 8 shows the monthly values of k_{stress} . The average monthly catchment water balance is seen in figure 2 and included is the water balance if water stress isn't considered in figure 3. It is apparent that if there is no water stress, $\Delta S \approx 0$ over the year. See [Appendix 1: Catchment Water Balance](#) For further plots relating to this.

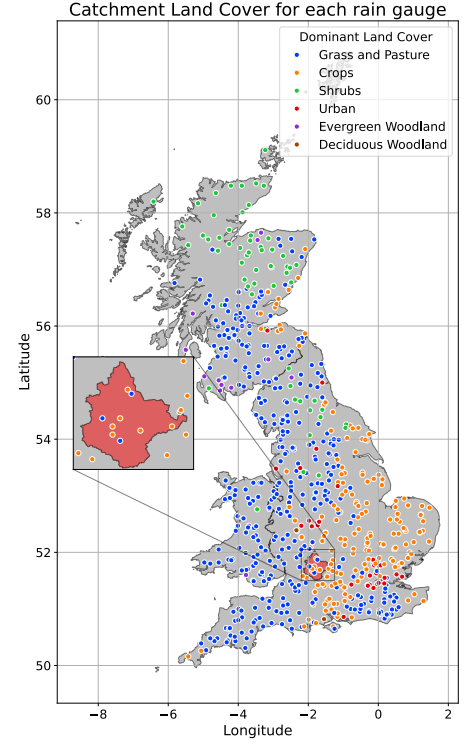


Figure 1: Dominant land cover
The average monthly values of k_{stress} The average monthly values of k_{stress} if water stress isn't considered in figure 3. It is apparent that if there is no water stress, $\Delta S \approx 0$ over the year. See [Appendix 1: Catchment Water Balance](#) For further plots relating to this.

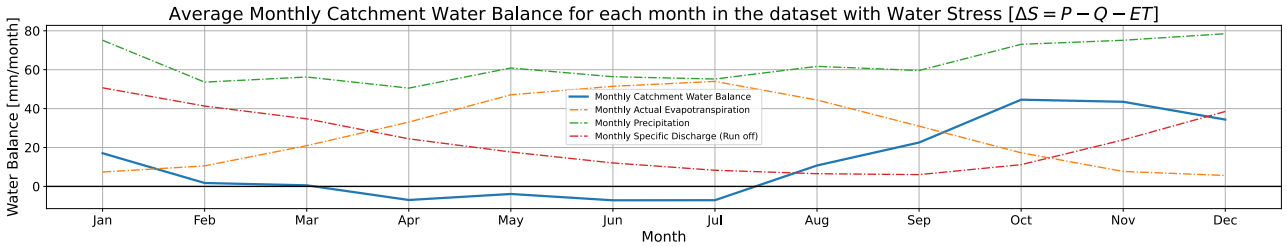


Figure 2: Monthly Average Water balance according to the data provided

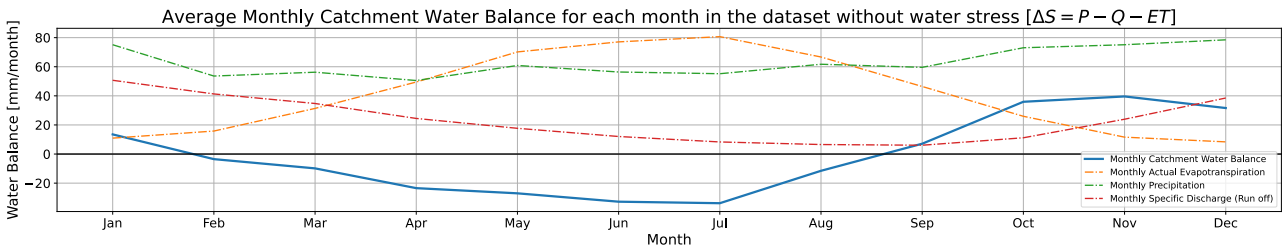


Figure 3: Monthly Average Water balance according to the data provided with no stress applied

2 Question 2: Annual Irrigation

The annual irrigation need is the difference between $PET_{catchment}$ and precipitation of the catchment. Figure 4 shows how the irrigation needs change over time, where the average yearly value is calculated to be **255.66mm a year**. Note that there is 0 irrigation need in 1990, when the region experienced significant floods.

3 Question 3: Model Calibration

The calibration of the model requires the use of an objective function to optimise a set of parameters that go into it. The objective function that was taken for this task was the Least Squares Method, which is as follows:

$$\min \sum_{i=1}^N (y - \hat{y})^2 \quad (2)$$

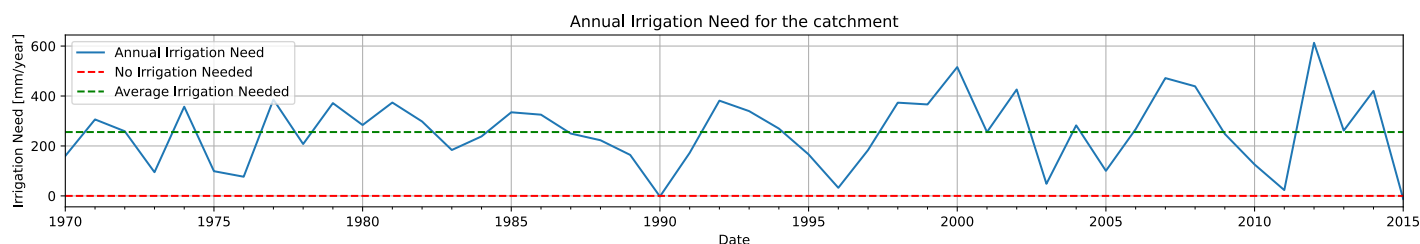


Figure 4: Irrigation needs across the dataset

This method was chosen as it provides a very interpretable fit statistic, and it is suitable for predictive models; 1 big error is more impactful than multiple small ones, this is ideal due to the stochastic nature of water modelling, where there is a high portion of irreducible error due to unobserved factors. Another suitable method would be the Nash-Sutcliffe Efficiency (NSE), where it uses a normalised scale $[\infty, 1]$ and values above 0.7 indicating a good fit. The data was split into an 80:20 train-test split to verify that the model is suitably fitted. Figure 5 shows the difference between the calibrated and uncalibrated model. Table 1 shows that the fit of the test portion of the model is even better than the entire calibration simulation, which indicates that the current parameters calculated are representative of the catchment. Note that the first year has been removed due to the initial warm-up stage as starting parameters become more representative.

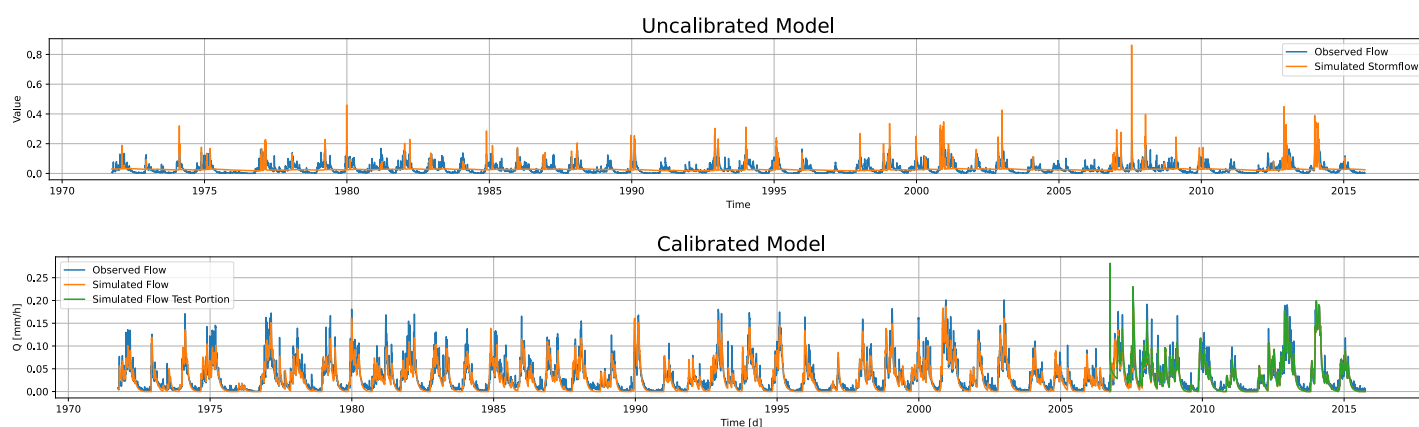


Figure 5: Uncalibrated and calibrated models

Table 1: Least Squares Values

Model	Uncalibrated	Calibrated	Calibrated (testing portion only)
Least Squares Value	13.19	2.60	0.86

4 Question 4: Groundwater Normalisation

There are 2 approaches taken with normalising the groundwater simulated. The first approach is normalising assuming that groundwater values are normally distributed. This clearly wouldn't yield the best results, as according to figure 15 in Appendix 4: Groundwater, the simulated groundwater is exponentially distributed. So to counter this, the values will be normalised using a Standardised Groundwater Index approach (SGI). Instead of fitting the values to a specific probability distribution, a non-parametric normal scores transform is applied to the monthly groundwater distribution, which assigns a probability to each value, ordered as such. This is then reordered back into its respective time series and made continuous, where SGI values below 0 indicate drought conditions (Bloomfield & Marchant 2013). In figure 6, around 1975/1976, the SGI for both the modelled and the actual groundwater SGI values indicated that the catchment was in extreme drought (≤ -2.0), which coincides with the actual drought that happened at the time. This shows that the current model can simulate changes in groundwater conditions pretty well, although the calculated Least Squares value of 286.76 indicates error is still significant. This might also be a testament to how difficult it is to measure groundwater values in the first place, where values are vaguely estimated, increasing the irreducible error due to unobserved factors.

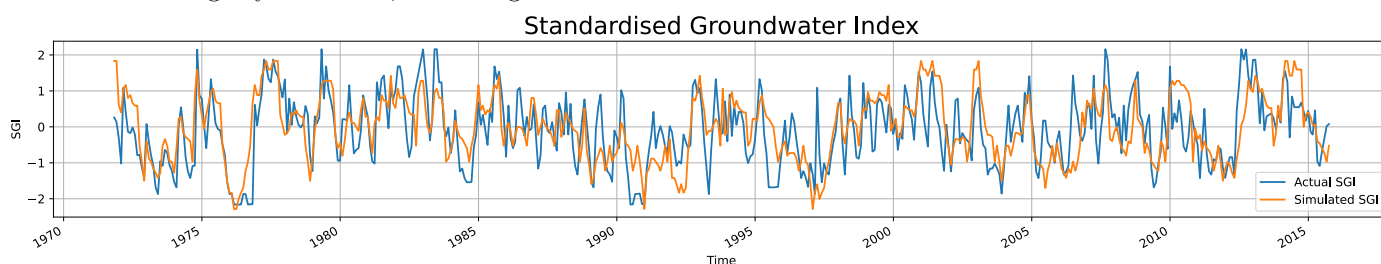


Figure 6: SGI values for the actual and simulated groundwater

5 Question 5: Reservoir Capacity

To calculate the Reservoir capacity, the Waitt curve method was employed. It was taken that the average daily demand is 142 litres (Water UK 2020), so the monthly demand was equivalent to 21300 m^3 per month, assuming it is constant. The abstracted flow must be aggregated monthly, to take seasonality to account. The Waitt curve can be determined using equation 3:

$$\min \left(\sum_{i=j}^{i=j+t-1} X_i \right) \quad (3)$$

Where t is the window size that is summed of the flow abstractions X_i in month i . The minimum value from a list of different window sizes is taken at the value of the Waitt curve. For example, if a window size of 3 months is taken, then the rolling sum of 3 months of the time series of flow abstraction is taken, then the minimum value out of that array is the value of the Waitt curve at 3 months. The reservoir size is the maximum difference between the demand curve and the Waitt curve, and the critical period is the window size it occurs at. This can be seen in figure 7, where the reservoir capacity is calculated as 0.0113 km^3 and the critical period as 55 months. This value is theoretically equivalent to the values obtained in the Rippl method, however it is less manual to compute. One downside to this method is that it can only account for historical droughts, and with climate change, bigger weather events may be more significant. It may be worth employing the Alexander method to calculate reservoir sizes for droughts of higher return periods.

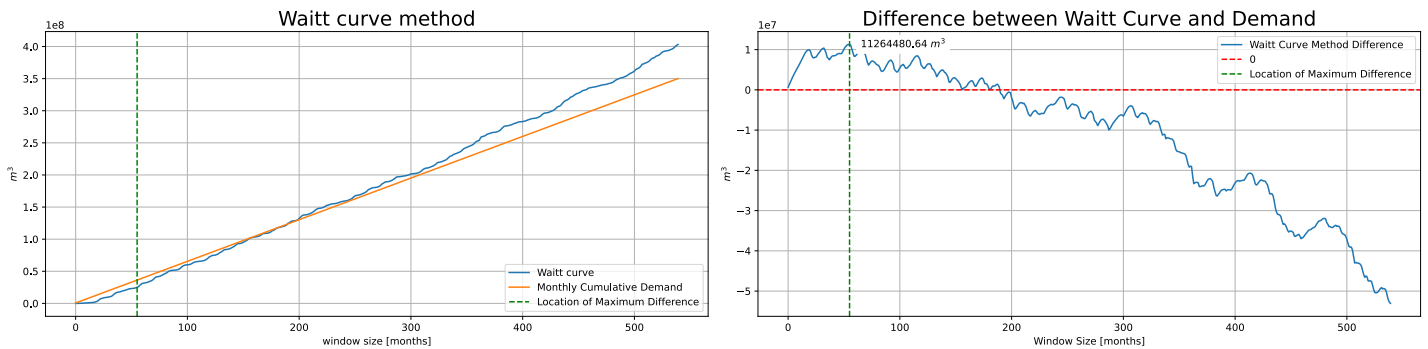


Figure 7: Waitt Curve method

6 Question 6

Strengths: The use of a lumped model provides a robust and computationally efficient way of predicting flow and ground-water levels with less data availability. It provides a good balance between realism and simplification to produce significant results that resemble real-life measurements. Automatic Calibration techniques such as Nelder-Mead are a good way of efficiently reaching local minima to optimise model parameters. The objective functions can be easily modified such that the error calculated can represent the model flexibility - for example, for a least-squares error, one big deviation from the training data has a larger error than multiple smaller errors, and other objective functions such as the NSE may take deviations differently. This allows for a degree of flexibility when it comes to optimisation. The methods used are less qualitative, so implementing an automatic feedback loop is much easier, and it allows for good supplementary information to make decisions based on the catchment. It was also seen that the models can detect significant events such as droughts (see 6) and floods, which shows that they may also be able to predict such events in the short term.

Weaknesses: Estimating a lumped model only gives a top-level understanding and does not capture interactions between different unique areas within the catchment. In addition, there is a certain point where the lumped model cannot exploit the available data anymore - this is where higher complexity physics-based models gain an advantage. Their feasibility is increasingly prevalent due to improvements in computational power over time. The estimation of the crop-specific PET has been done quite haphazardly, assuming crop cycles stay constant over time, this is not the case. This is also an issue for the land cover and the calibrated model parameters, where they are likely to experience change over time. That is why water resources are so hard to predict, as it is difficult to determine what anthropomorphic and natural changes will occur. To counter this, an uncertainty to parameters should be added, as the water cycle is a naturally stochastic process. Also, when calibrating the model, only 1 objective was used, so this might produce a suboptimal model, not even considering that calibration may also optimise into a local instead of a global minimum. Both Genetic algorithms and multi-objective optimisation can help counter these problems. To add to problems of uncertainty, predictions of the reservoir volumes were taken to be deterministic, using previous flow and constant demand value. To improve this, demand should be predicted according to future conditions, and employment of the Alexander method should be done to examine the reservoir volume for different scenarios such as a 100-year flood. Finally, discrepancies between data values have been a challenge, where conflicting values complicates calculation accuracy. For example, 2 different sets of land cover values can be obtained from CAMELS and the National River Flow Archive, this increases the ambiguity of the results obtained. Ambiguity seems to be a common trait of water resources analyses, which should always be considered carefully.

References

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- Coxon, G., Addor, N., Bloomfield, J. P., Freer, J., Fry, M., Hannaford, J., Howden, N. J. K., Lane, R., Lewis, M., Robinson, E. L., Wagener, T. & Woods, R. (2020), 'Camels-gb: hydrometeorological time series and landscape attributes for 671 catchments in great britain', *Earth System Science Data* **12**, 2459–2483.
- Defra (2024), 'Agricultural facts: South east region'.
URL: <https://www.gov.uk/government/statistics/agricultural-facts-england-regional-profiles/agricultural-facts-south-east-region>
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URL: <https://www.water.org.uk/news-views-publications/news/vast-majority-brits-have-no-idea-how-much-water-they-use-each-day>

7 Appendices

7.1 Appendix 1: Catchment Water Balance

Figure 8: Monthly water stress values

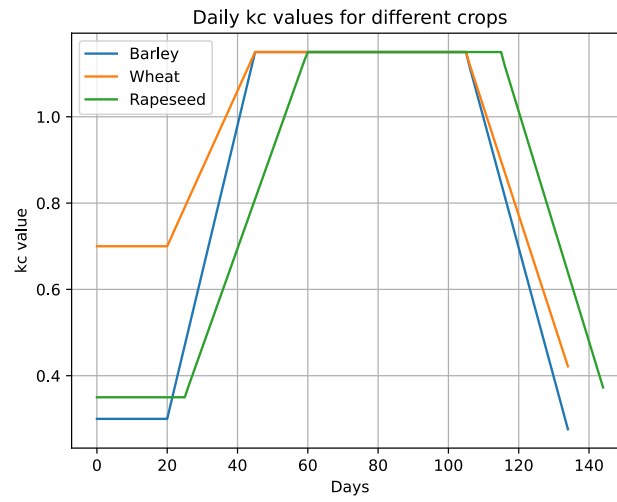


Figure 9: k_c for all crop types

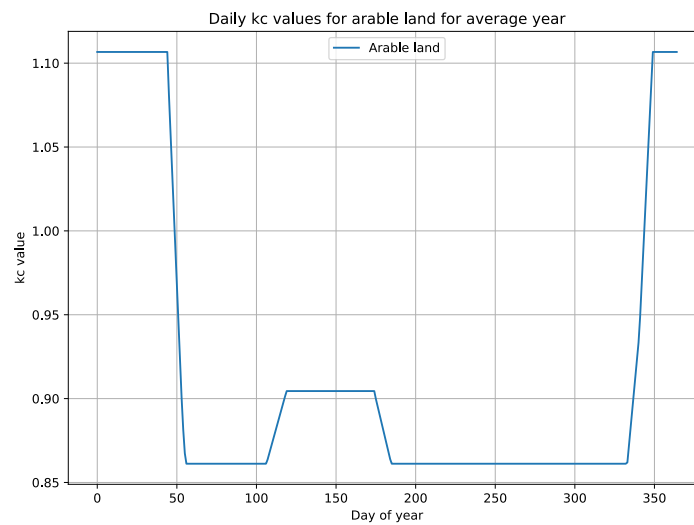


Figure 10: $k_c^{(arable)}$ for arable plants

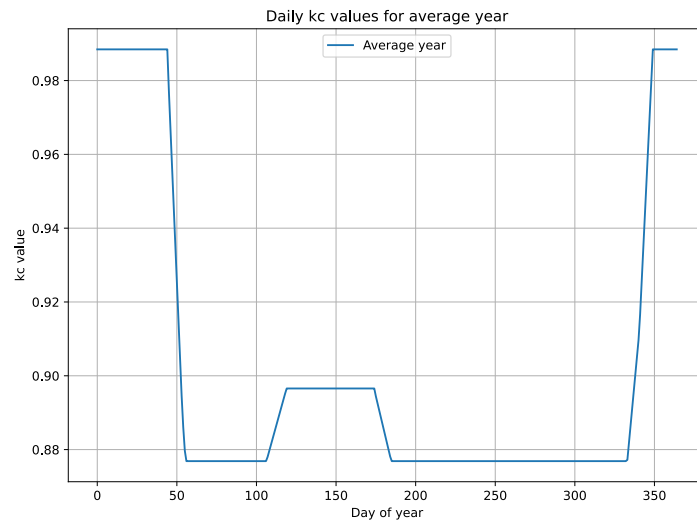


Figure 11: Monthly Average Water balance according to the data provided

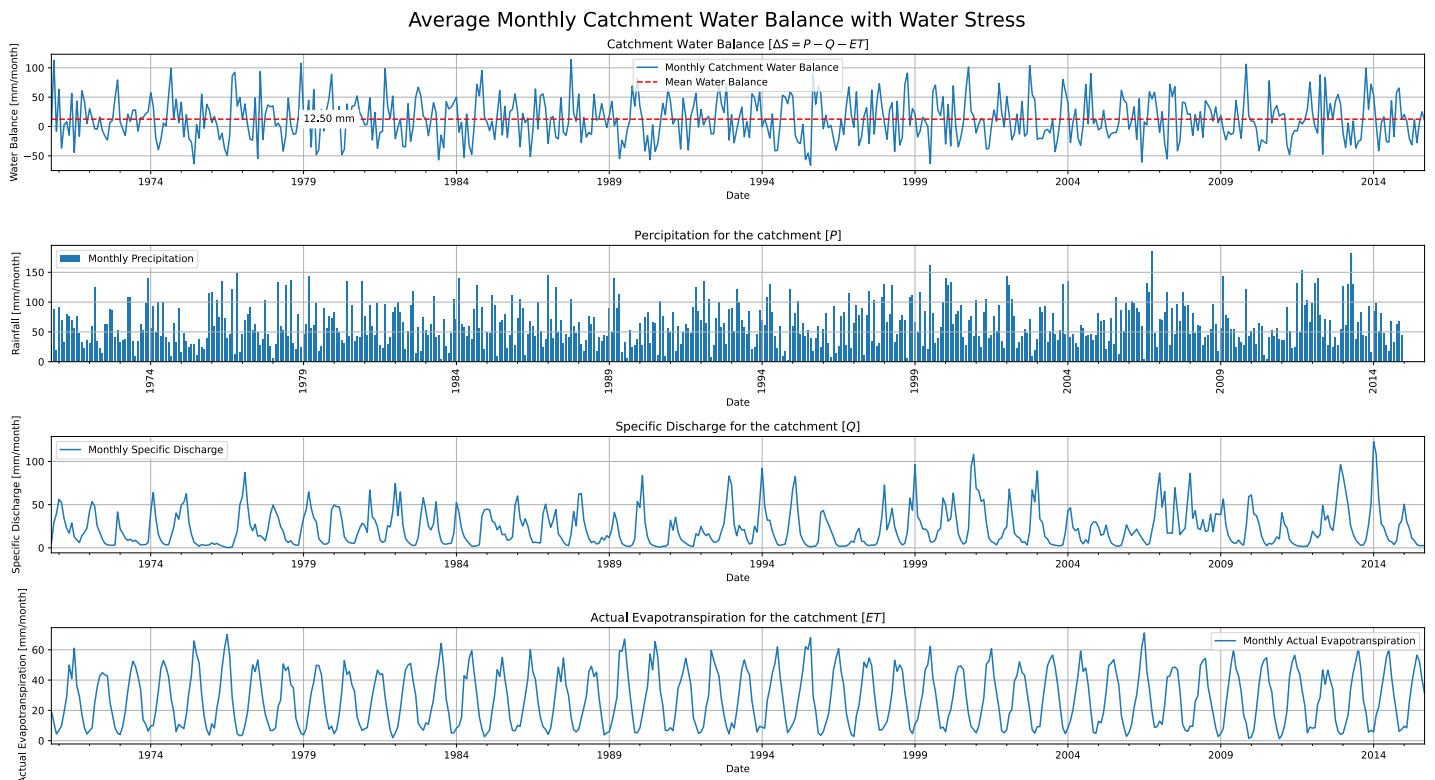


Figure 12: Water Balance across the Dataset

7.2 Appendix 4: Groundwater

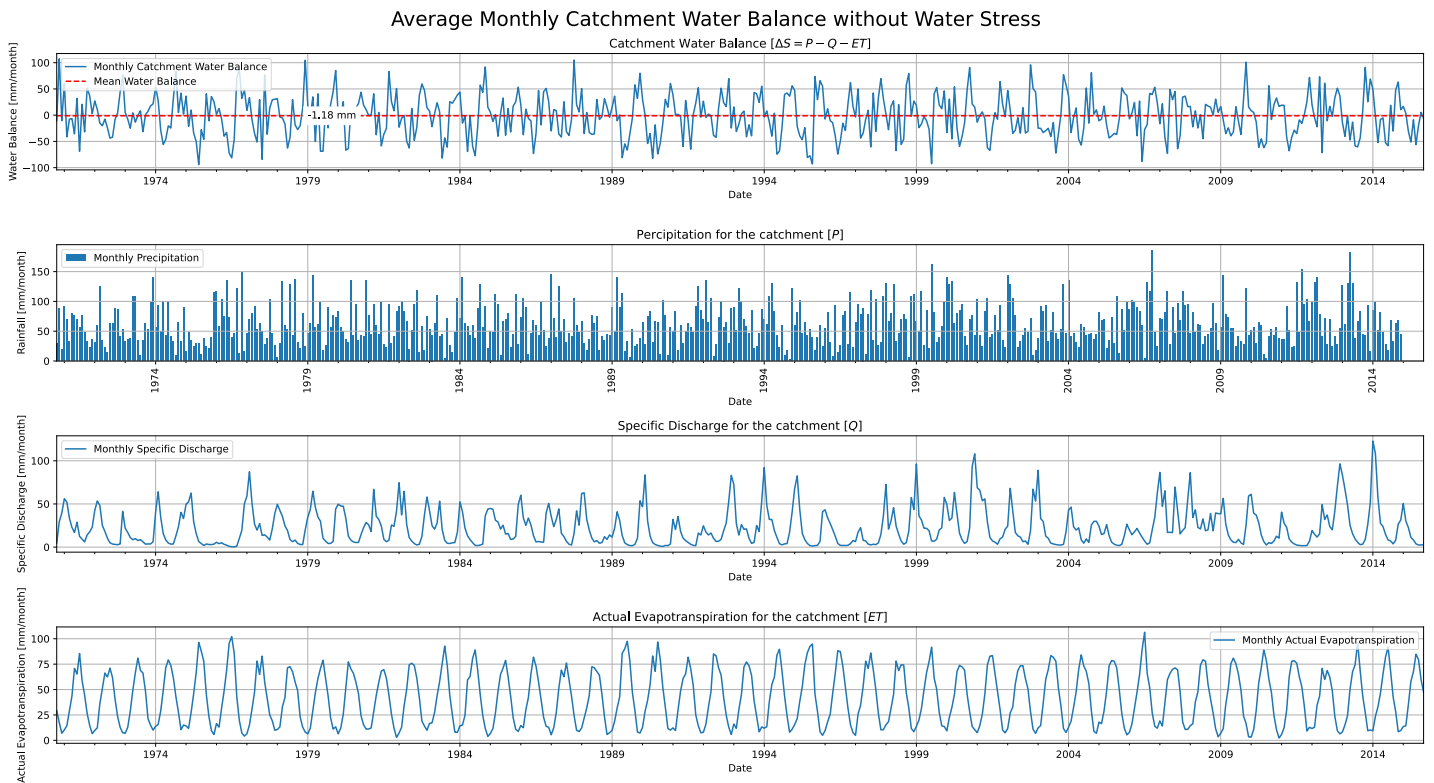


Figure 13: Water Balance across the Dataset without water stress

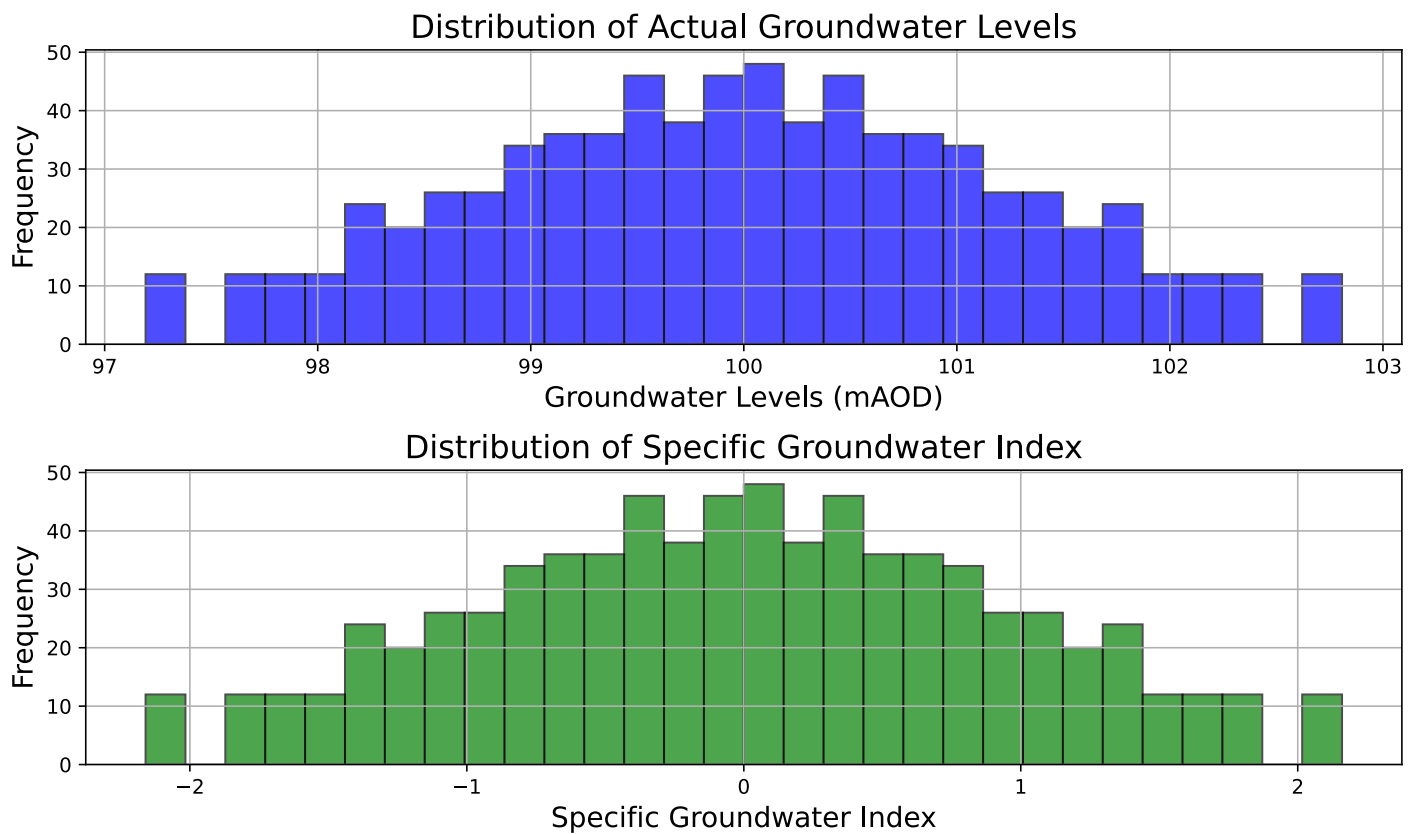


Figure 14: Actual Groundwater Distributions

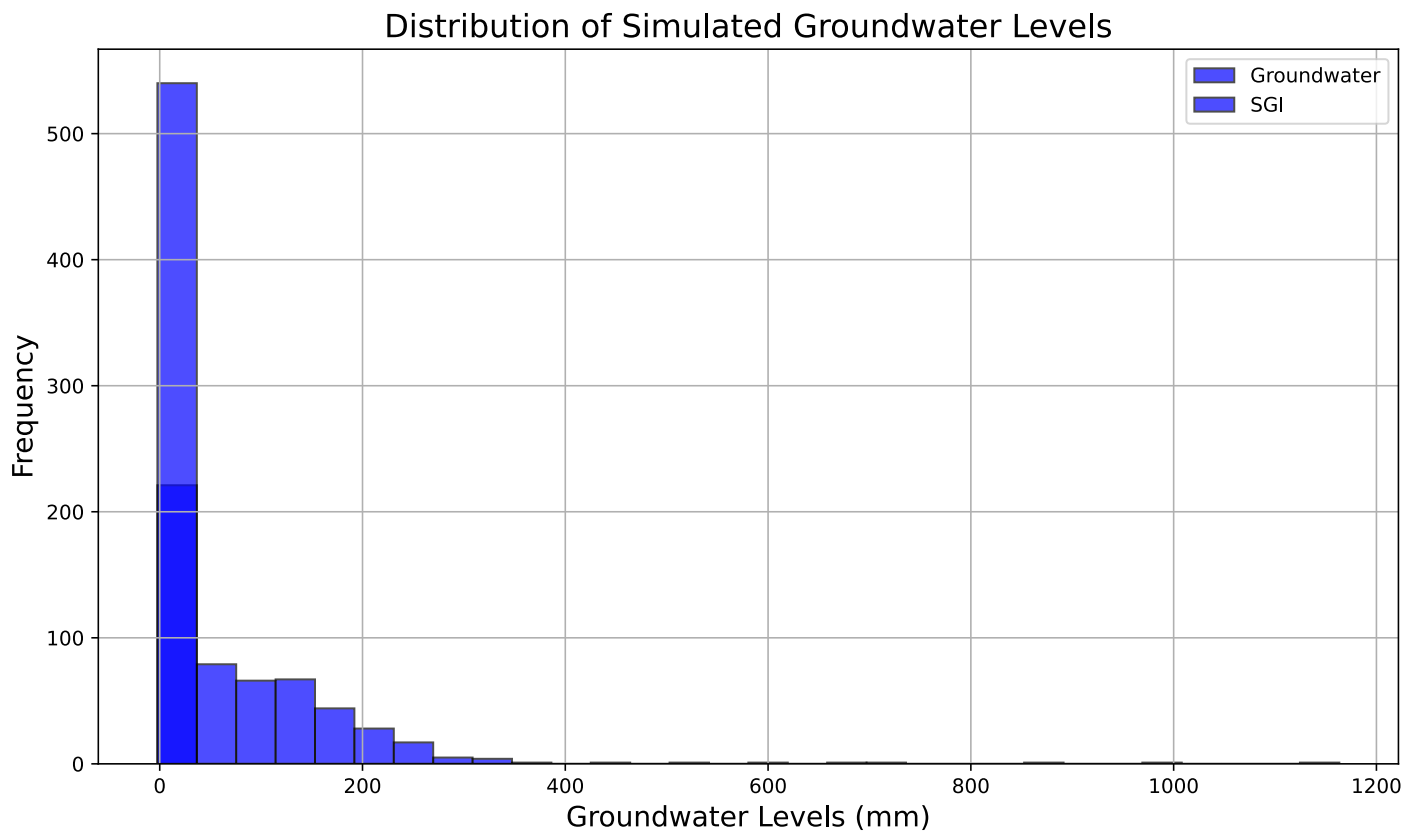


Figure 15: Simulated groundwater distribution

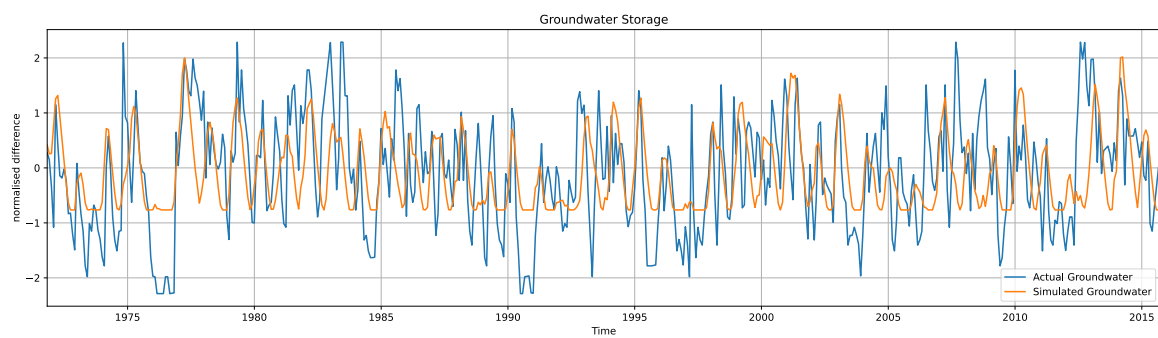


Figure 16: Groundwater using Normal Distribution normalisation